

Question Paper

Exam Date & Time: 30-Apr-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. DEGREE EXAMINATIONS - APRIL/MAY 2025

SUBJECT: MAT 1271/MAT_1271 - ENGINEERING MATHEMATICS - II

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Find the maxima and minima of the function (4)

$$f(x, y) = x^3 + y^3 - 12xy.$$

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1B) If $Z = \frac{\cos y}{x}$ and $x = u^2 - v, y = e^v$. (3)

Find $\frac{\partial z}{\partial u}$ and $\frac{\partial z}{\partial v}$.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) If $u(x, y) = \log_e \left(\frac{x^4 + y^4}{x + y} \right)$, then show that (3)

$$x^2 \frac{\partial^2 u}{\partial x^2} + 2xy \frac{\partial^2 u}{\partial x \partial y} + y^2 \frac{\partial^2 u}{\partial y^2} = -3.$$

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) Show that the plane $2x - y - 2z = 16$ touches the sphere (4)
 $x^2 + y^2 + z^2 - 4x + 2y + 2z - 3 = 0$ and hence find the point of contact.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2B) Find the Taylor's series expansion of (3)

$$f(x, y) = (1 + x - y)^{-1} \text{ about the point } (1, 1)$$

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) (3)

Evaluate $\lim_{x \rightarrow 1} (x^2 - 1) \tan \frac{\pi x}{2}$

Course Outcome: "CO2" / Cognitive Level: "Application"

3A) Test the nature of the series (4)

$$\sum_{n=1}^{\infty} \frac{((2n+1)x)^n}{n^{n+1}}, \quad x > 0$$

Course Outcome: "CO6" / Cognitive Level: "Application"

3B) Change the order of integration and evaluate (3)

$$\int_0^1 \int_x^{2x} (x^2 + y^2) dy dx.$$

Course Outcome: "CO4" / Cognitive Level: "Analysis"

3C) Use double integrals to find the area of the region bounded by the curves (3)

$$y = 2 - x^2 \text{ and } y = x.$$

Course Outcome: "CO4" / Cognitive Level: "Application"

4A) Use Laplace transform to solve the initial value problem (4)

$$y'' + 2y' + 5y = e^{-t} \sin t \quad \text{given} \\ y(0) = 0, \quad y'(0) = 1$$

Course Outcome: "CO5" / Cognitive Level: "Application"

4B) Express the function $f(t) = \begin{cases} t^2, & 0 < t \leq 1 \\ 4t, & t > 1 \end{cases}$ in terms of unit step (3)

function and hence find its Laplace transform.

Course Outcome: "CO5" / Cognitive Level: "Application"

4C) Find the Laplace transform of $(3t^5 + 4e^{-5t} - 3 \sin 6t)e^{2t}$. (3)

Course Outcome: "CO5" / Cognitive Level: "Application"

5A) Test the convergence of the series $\sum_{n=1}^{\infty} \frac{(x+2)^n}{3^n n}$ (4)

Course Outcome: "CO6" / Cognitive Level: "Analysis"

5B) Use triple integrals to find the volume bounded by (3)

$$x^2 + y^2 + z = 4, \quad x^2 + y^2 = 4y \text{ and } z = 0.$$

Course Outcome: "CO4" / Cognitive Level: "Application"

5C) (3)

Use Beta and Gamma functions to find the value of

$$\int_0^{\infty} \sqrt{x} e^{-x^2} dx \times \int_0^{\infty} \frac{1}{\sqrt{x}} e^{-x^2} dx.$$

Course Outcome: "CO4" / Cognitive Level: "Application"

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Event Name			Subject
MIT MPL and BLR- BTech-M Sc-MCA - II-IV and VI Sem - End Semester Examination - Apr-May 2025			MAT- 1271:Engineering Mathematics- II
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	6
3	3	CO 3	4
4	4	CO 4	12
5	5	CO 5	10
6	6	CO 6	8
Total			50

* **NA** - CO Not Selected

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MIT MPL and BLR- BTech-M Sc-MCA - II-IV and VI Sem - End Semester Examination - Apr-May 2025			MAT- 1271:Engineering Mathematics- II
#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	12	39(78%)
4	Analysis	3	11(22%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)